

Question Paper

Exam Date & Time: 01-Jan-2026 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026

SUBJECT: ECE 2123/ECE_2123 SIGNALS & SYSTEMS

(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Use the frequency- differentiation property to find the FT of $x(t)=t e^{at} u(-t)$ (5)

$$\text{Given FT of } e^{at} u(-t) \longleftrightarrow \frac{1}{(-j\omega + a)}$$

(FT - Fourier Transform)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Determine the Fourier Series(FS) coefficients and the FS representation for the given signal (3)

$$x(t) = 3 + 2\cos(4\pi t) + \sin(6\pi t)$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Determine whether the signals are energy or power signals (2)
- i) A square wave with amplitude $A=2$ and period $T=0.4s$
 - ii) A rectangular pulse of amplitude $A=10$ and duration $T_1=5\mu s$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Determine whether the following systems are a) Linear b) time invariant c) stable (5)

i) $y(t) = e^{tx(t)}$

ii) $y(t) = tx(t)$

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) Sketch direct form-I & II structures for the system described by the differential equation. (3)

$$\frac{d^3 y(t)}{dt^3} + \frac{2dy(t)}{dt} + 3y(t) = x(t) + \frac{3dx(t)}{dt}$$

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Given the transfer function: (2)

$$H(z) = \frac{1 - z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$$

Sketch the pole-zero diagram of the transfer function and identify the region of convergence (ROC) that guarantees the system is both causal and stable. Then, compute the system's impulse response.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Using the inverse DTFT, determine the time domain signal for the spectrum. (5)

$$X(\omega) = \begin{cases} 1, & |\omega| \leq B_0 \\ 0, & B_0 \leq |\omega| < \pi \end{cases}$$

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3B) Find the DTFS representation for the Discrete time signal (3)

$$x[n] = \cos\left(\frac{n\pi}{30}\right) + 2\sin\left(\frac{n\pi}{90}\right)$$

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) Determine whether the signal $x[n] = 0.8^n u[n]$ is an energy signal or signal? Calculate its energy or power as appropriate. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) Determine if the following system are causal, linear and time-invariant. $x[n]$ is the input and $y[n]$ is the output. (5)

a) $y[n] = x[n]x[n-2]$

b) $y[n] = a^n x[n]$

c) $y[n] = \log_{10} |x[n]|$

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) Using properties of the z-transform, find the z-transform of the following sequences (3)

a) $x[n] = u[-n]$

b) $x[n] = 2^n u[n-2]$

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) Plot the pole-zero pattern and determine if the following system is stable (2)

$$y[n] = y[n - 1] - 0.8y[n - 2] + x[n] + x[n - 2]$$

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5A) Define Nyquist Sampling rate. Determine the Nyquist sampling rate for (5)

$$x(t) = \sin \frac{4\pi t}{\pi t}.$$

With the help of a suitable plot, compare the two scenarios:

- a) When the signal is sampled below Nyquist rate and
b) When the signal is sampled above the Nyquist rate.

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) Evaluate the linearity and causality properties of the following systems (3)

a) $y[n] = n^2 x[n]$

b) $y[n] = x[2n]$

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5C) Consider a signal $x(t) = \cos(\pi t)$. (2)

Sketch the frequency spectrum of the signal when sampled at three different sampling intervals:

- a) $T_s = 1/4$,
b) $T_s = 1$

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_2123:SIGNALS AND SYSTEMS

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	13
5	5	CO 5	7
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_2123:SIGNALS AND SYSTEMS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	11	36(72%)
4	Analysis	4	14(28%)
5	Create	0	0(0%)

6	Evaluate	0	0(0%)
	Total	15	50 (100%)